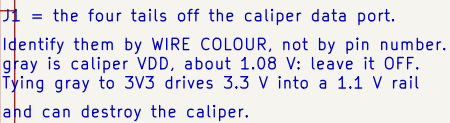


Two inverting common-emitter shifters lift the caliper's 0–1.1 V logic to 0–3.3 V. Caliper high → transistor on → ESP32 pin pulled LOW. That is SHIFTER_INVERTS 1 in config.h.



BC547 pinout: flat face toward you, legs down,
left to right = C B E.
A 2N3904 is E B C, the mirror image. If a stage
will not switch, swap the outer two legs.

U1 is a classic ESP32 DevKit (WROOM-32).
Pin names match the board silkscreen.
The X marks are pins the DevKit already
handles for you -- do not wire them.
3V3 comes from the DevKit regulator, over USB.

SW1/SW2 are for BLE mode only -- not needed for CALIPER_SNIFFER_MODE bring-up.

Sheet: /		D
File: protoboard.kicad_sch		
Title: BLE Digital Caliper -- protoboard prototype		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.4		Id: 1/1